

Developer Experience in the AI Era: A Power BI Analysis of the Stack Overflow Developer Survey 2025

Project Overview

This project analyzes the Stack Overflow Developer Survey 2025 to understand developer profiles, AI adoption, attitudes toward AI, job satisfaction, compensation, and technology usage. The goal is to build an interactive Power BI report that shows how developer background and experience relate to AI learning, AI-tool usage, and workplace outcomes.

Objective

The main objective of this project is to explore how developers are adapting to AI in their work and learning environments, and how this relates to factors such as role type, education, experience, job satisfaction, compensation, and technology preferences.

Why this dataset was selected

This dataset was chosen because it is a large real-world survey dataset with rich information on developer demographics, employment, education, work experience, AI-related behavior, compensation, and tools used. It is also realistic and challenging because it contains missing values and multi-valued fields, making it more suitable for practical Power BI analysis than a small pre-cleaned dataset.

Main themes of analysis

This project focuses on the following themes:

- Respondent and workforce profile
- AI learning and adoption
- AI sentiment, trust, and perception of complex-task handling
- Job satisfaction and compensation
- programming language and AI model usage

Dataset source

Source: Stack Overflow Developer Survey 2025

Link: https://survey.stackoverflow.co/?utm_sour

Scope of the project

Instead of using all columns from the original survey, this project works with a selected subset of relevant fields to keep the analysis focused, meaningful, and manageable. Additional transformations were performed for multi-valued columns such as programming languages and AI models to support relationship-based analysis in Power BI.

Expected outcome

The final report is designed to answer key questions about who the survey respondents are, how they are adopting AI, how they perceive AI tools, and how these patterns connect to outcomes such as job satisfaction, compensation, and technology usage.

Table Name	Key Columns & Data Types
AIModels (Dimension Table)	RespondId (Whole Number) - respondent ID; AIModel (Text) - AI model used.
Languages (Dimension Table)	RespondId (Whole Number) - respondent ID; Language (Text) - programming language used.
MainRespondents (Fact Table)	RespondId (Whole Number) - unique respondent identifier; Age (Text) - age group of the respondent; EdLevel (Text) - highest education level completed; Employment (Text) - c WorkExp (Whole Number) - years of professional work experience; YearsCode (Whole Number) - years of coding experience; DevType (Text) - primary developer role; RemoteWc such as remote, hybrid, or in-person; Industry (Text) - industry sector of the respondent; Country (Text) - respondent country; LearnCodeAI (Text) - whether and how the respond LearnCodeAI_YesNo (Text) - simplified AI-learning grouping; AIThreat (Text) - whether the respondent sees AI as a threat; AISent (Text) - detailed sentiment toward AI; AISentime sentiment category; AIAcc (Text) - trust in AI accuracy; AIComplex (Text) - perception of AI ability on complex tasks; AIAgents (Text) - detailed AI agent usage category; AIAgentsl agent usage grouping; ExperienceBand (Text) - grouped work experience category; YearsCodingBand (Text) - grouped coding experience category; ConvertedCompYearly (Whole compensation in standardized currency; JobSat (Whole Number) - job satisfaction score.

Started with the original Stack Overflow Developer Survey 2025 dataset containing 49,191 rows and 172 columns. From this full survey, a focused subset of 21 columns was selected to match the project themes of developer profile, AI adoption, AI attitudes, compensation, job satisfaction, and technology usage.

Split the two original multi-valued fields into separate analysis tables so they could be modeled correctly in Power BI. LanguageHaveWorkedWith was transformed into a Languages table with Responseld and Language, and AIModelsHaveWorkedWith was transformed into an AIModels table with Responseld and AIModel.

Built a 3-table model with MainRespondents as the central fact table and Languages and AIModels as related dimension tables. Relationships were created using Responseld, allowing respondent-level analysis to connect with language usage and AI-model usage

Cleaned categorical columns by standardizing missing and inconsistent values. Blank responses in selected text fields were replaced with **No Response** so that missing categories remained visible in the analysis instead of being silently dropped.

Simplified several long survey labels for readability in report visuals. For example, values such as **Other (please specify)** were shortened to **Other**, and long education, remote-work, and AI-related labels were standardized into cleaner reporting categories.

Preserved missing values in numeric columns such as WorkExp, YearsCode, ConvertedCompYearly, and JobSat as **null** rather than replacing them with zero. This avoided distorting averages, medians, and other summary statistics.

Applied a row-level usefulness filter to remove records that did not contribute meaningful information to the selected project scope. This step removed **852 non-informative rows**, leaving a final MainRespondents table with **48,339 rows**.

Created several derived grouping and classification fields to make the survey data more interpretable in dashboards and comparisons. These included ExperienceBand, YearsCodingBand, JobSatisfactionBand, AISentimentGroup, Compensation Band, Compensation Capped, LearnCodeAI_YesNo, and AIAgentsUse.

Grouped detailed AI-related responses into more analytical categories. For example, detailed sentiment values were collapsed into **Positive**, **Neutral**, and **Negative**, AI-learning responses were simplified into **Yes/No**, and AI-agent usage was grouped into cleaner reporting categories.

Built an additional flow table for the Sankey visual by filtering valid LearnCodeAI_YesNo, AIAgents, and AISentimentGroup values, grouping them into flow counts, and appending them into one Source–Destination–Value structure.

Q1:

Who are the survey respondents in terms of developer role, employment status, education level, experience, and geographic distribution?

Data Relationships Explored:

This question mainly explores **nominal comparison** and **ranking** relationships.

- **Nominal comparison** is used to compare respondent counts across categories such as developer role, employment status, education level, and country.
- **Ranking** is used to order the largest respondent groups, especially for developer roles and top countries.

First Chart: Developer Role Distribution

Displays the number of respondents across developer roles (DevType) using a bar chart to compare which roles are most represented.

Second Chart: Employment Status Distribution

Shows how respondents are distributed across employment categories such as employed, self-employed, student, and not employed.

Third Chart: Education Level Distribution

Summarizes the respondent base by education level to compare the strongest academic groups.

Fourth Chart: Country Distribution (Top Countries)

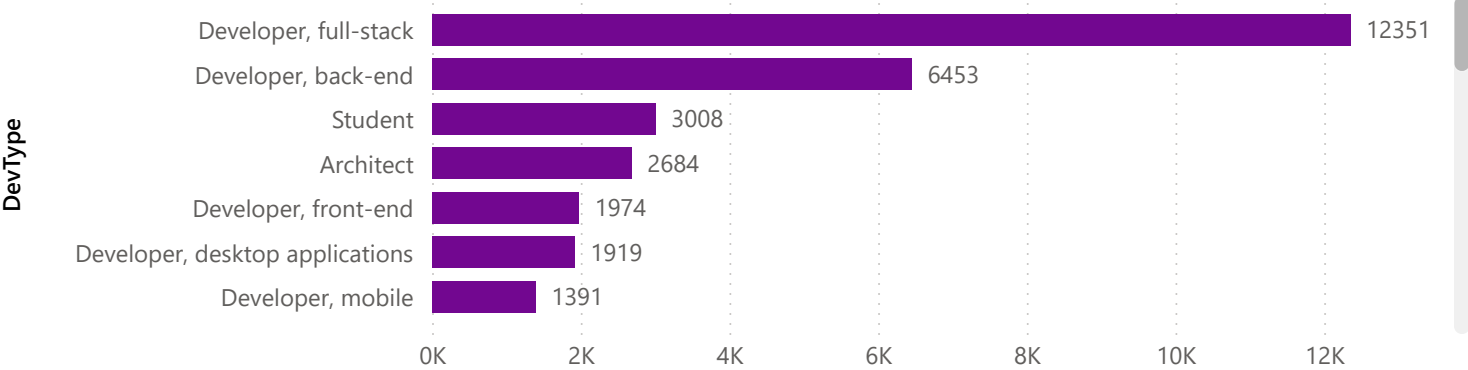
Ranks the main respondent countries to show the strongest geographic concentrations in the survey sample.

KPI Cards:

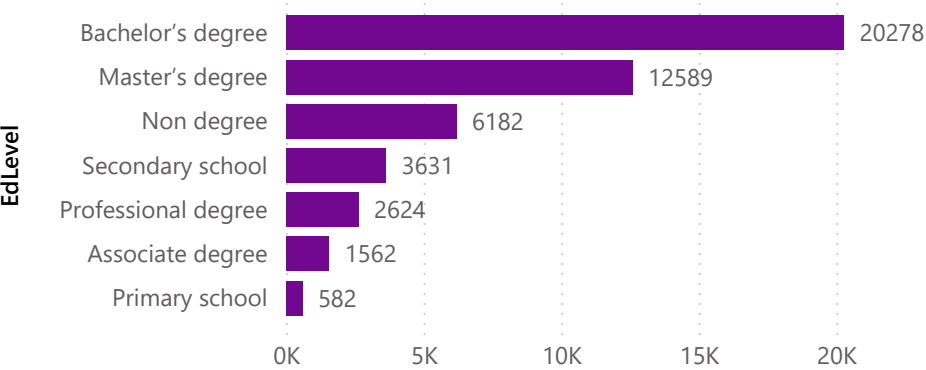
TotalRespondents, AvgWorkExperience, and AvgYearsCoding provide a summary of the respondent base

RESPONDENT PROFILE

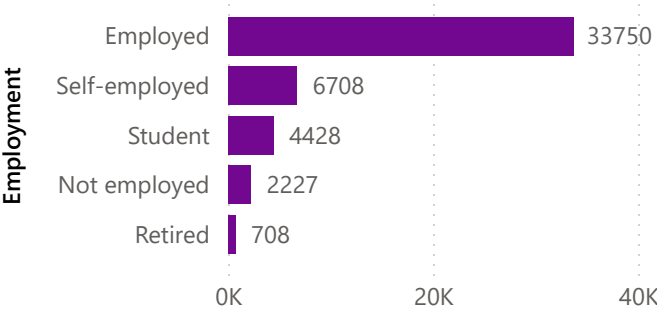
TotalRespondents by DevType



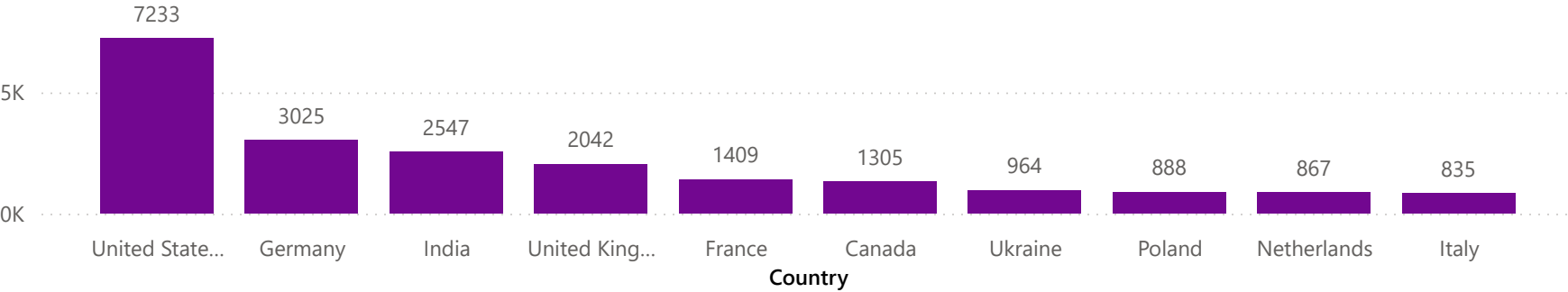
TotalRespondents by EdLevel



TotalRespondents by Employment



TotalRespondents by Country



48339

TotalRespondents

13.37

AvgWorkExperience

16.57

AvgYearsCoding

RemoteWork

Select all

Flexible

Hybrid

In-person

Remote

Age

All

The respondent profile shows that the survey is heavily concentrated in mainstream software development roles, with **full-stack developers** forming the largest group and **back-end developers** consistently appearing as the second strongest role.

Employment patterns show that most respondents are **actively employed**, while self-employed, student, and not-employed groups are much smaller.

The education distribution is also clearly concentrated in **bachelor's** and **master's** degree holders, indicating that the respondent base is academically strong.

Across the age filters, the profile changes in a meaningful way: the **18–24** group contains a stronger student presence and much lower average experience, while the **25–34** and **35–44** groups form the core of the professional sample. As age increases, both **average work experience** and **average years coding** rise steadily, showing a realistic career progression pattern.

Geographically, the **United States** remains the largest respondent country overall, with **Germany** and **India** also contributing strongly across several views.

Insight: This means the dataset is not a uniform sample of all possible developers, but is instead centered on **professionally active, mid-career software developers** with relatively strong educational backgrounds. As a result, the later findings on AI adoption, compensation, and technology usage should be interpreted in the context of a respondent base that is more workforce-oriented and more heavily represented by established development roles than by beginners or non-technical participants.

Q2 :

How are developers learning AI, and how do AI sentiment, AI threat perception, and perceived AI capability differ across respondents?

Data Relationships Explored:

This question mainly explores **part-to-whole**, **nominal comparison**, and **ranking** relationships.

- **Part-to-whole** is used in the AI-learning pie chart to show how the respondent base is divided across AI-learning categories.
- **Nominal comparison** is used to compare AI capability ratings, AI threat perception by employment status, and sentiment composition across developer roles.
- **Ranking** is used to order the most represented developer roles in the sentiment comparison chart.

First Chart: AI Learning Distribution

Displays how respondents are distributed across AI-learning categories, including learning AI for job/career purposes, personal interest, non-AI learning, or no learning in the past year.

Second Chart: AI Complexity Perception

Shows how respondents rate AI's ability to handle complex tasks, helping identify whether AI is seen as capable, limited, or not widely used for harder tasks.

Third and Fourth Charts: KPI Cards for % Learned AI and % Using AI Agents

Summarize the share of respondents who learned AI and the share who actively use AI agents, allowing quick comparison across different sentiment groups.

Fifth Chart: AI Sentiment Mix by Developer Type

Compares the percentage distribution of positive, neutral, and negative AI sentiment across major developer roles.

Sixth Chart: Employment by AI Threat Perception

Displays the percentage split of respondents who do or do not see AI as a threat across employment categories.

Slicer: AISentimentGroup

Allows comparison across **Positive**, **Neutral**, and **Negative** AI sentiment groups.

AI ADOPTION AND PERCEPTION

Sentiment Towards AI

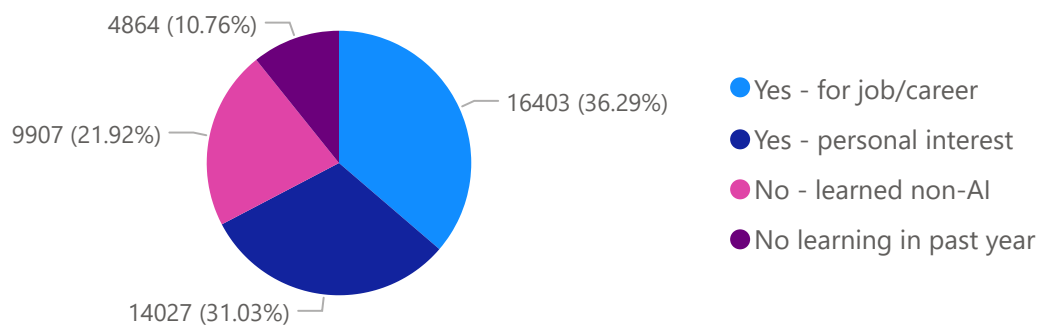
Deselect all

Negative

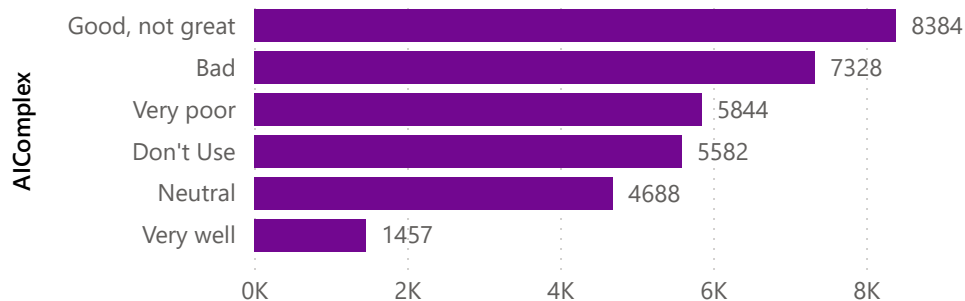
Neutral

Positive

Total Respondents by LearnCodeAI



Respondents Opinion on AI for complex tasks



0.67

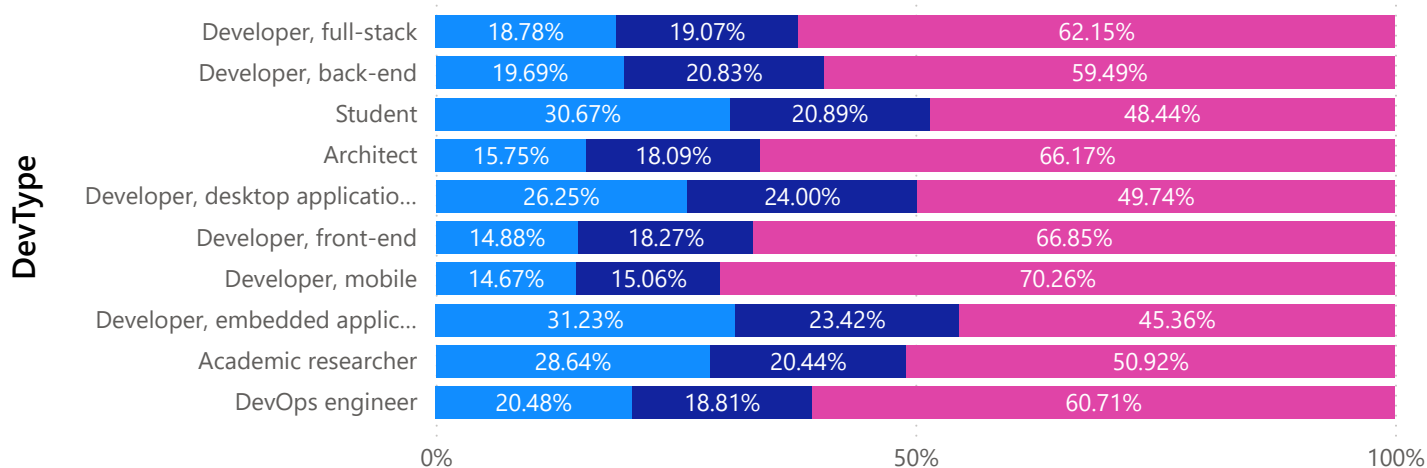
% Learned AI

0.31

% Using AI Agents

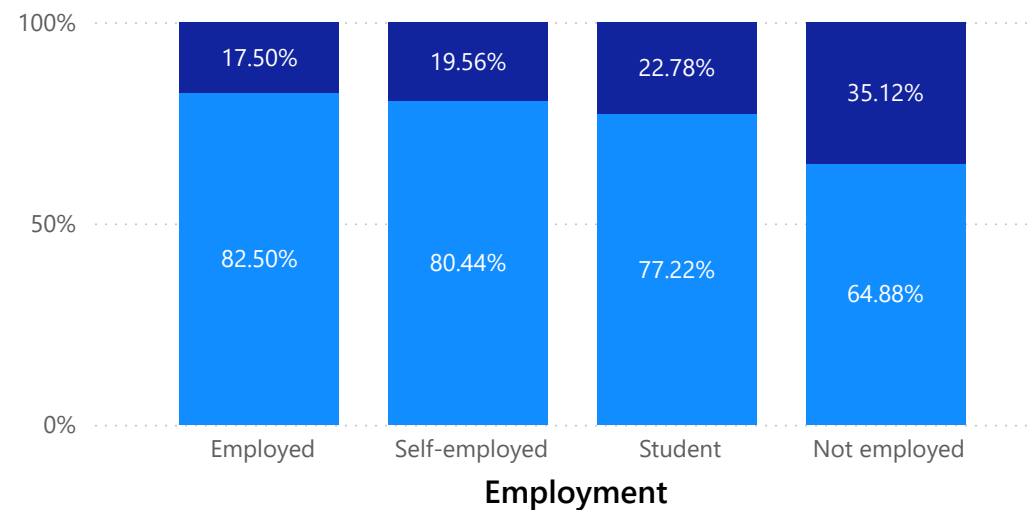
Top 10 Total Respondents by Developer Type

● Negative ● Neutral ● Positive



Total Respondents by Employment and AI Threat

● No ● Yes



AI learning is already widespread across the respondent base, but it varies sharply by sentiment group. In the full sample, **67%** of respondents learned AI and **31%** report using AI agents. The strongest learning categories are AI for **job/career** and **personal interest**, while non-AI learning and no recent learning are smaller. This gap becomes much clearer by sentiment: in the **positive** group, **83%** learned AI and **42%** use AI agents, compared with **55%** and **21%** in the **neutral** group, and only **29%** and **9%** in the **negative** group. AI capability ratings also decline as sentiment becomes more negative. Most respondents rate AI as “**Good, not great**” rather than “**Very well**,” suggesting cautious rather than full confidence. Role-based patterns are also uneven, with stronger positive shares in roles such as **mobile developer, front-end developer, architect, and DevOps engineer**, while roles such as **student, embedded applications, and desktop applications** appear more mixed. The AI-threat chart shows that most respondents still do **not** see AI as a threat, but threat perception is highest among the **not employed** and **student** groups.

Insight: The strongest pattern is that **AI learning and AI-agent usage rise together with positive AI sentiment**, while lower AI exposure is associated with more skepticism, lower perceived capability, and higher threat perception. Even so, the most common positive view is still “**Good, not great**,” which suggests that AI adoption is growing faster than full confidence in AI’s ability to handle complex work.

Q3:

How do compensation and job satisfaction vary across AI sentiment, AI-agent usage, developer role, and coding experience?

Data Relationships Explored:

This question mainly explores **part-to-whole**, **nominal comparison**, and **ranking** relationships, with a limited outcome comparison across experience groups.

- **Part-to-whole** is used in the 100% stacked bar chart to show how AI sentiment is distributed within each compensation band.
- **Nominal comparison** is used to compare average job satisfaction by AI-agent usage and median compensation across developer roles.
- **Ranking** is used to order developer roles by compensation levels

First Chart: AI Sentiment Distribution by Compensation Band

Displays the percentage distribution of positive, neutral, and negative AI sentiment across compensation bands.

Second Chart: Average Job Satisfaction by AI-Agent Usage

Compares average job satisfaction between respondents who use AI agents and those who do not.

Third Chart: Median Compensation by Developer Type and AI Sentiment Group

Shows how median compensation differs across major developer roles when split by AI sentiment group.

Fourth Chart: Compensation by Experience Level and AI Sentiment Group

Provides a supporting comparison of compensation patterns across experience groups and sentiment categories.

KPI Cards:

AvgJobSatisfaction and MedianCompensation summarize the overall outcomes for the selected group.

Slicer: YearsCodingBand

Allows compensation and job satisfaction patterns to be compared across coding-experience stages.

Compensation and Satisfaction

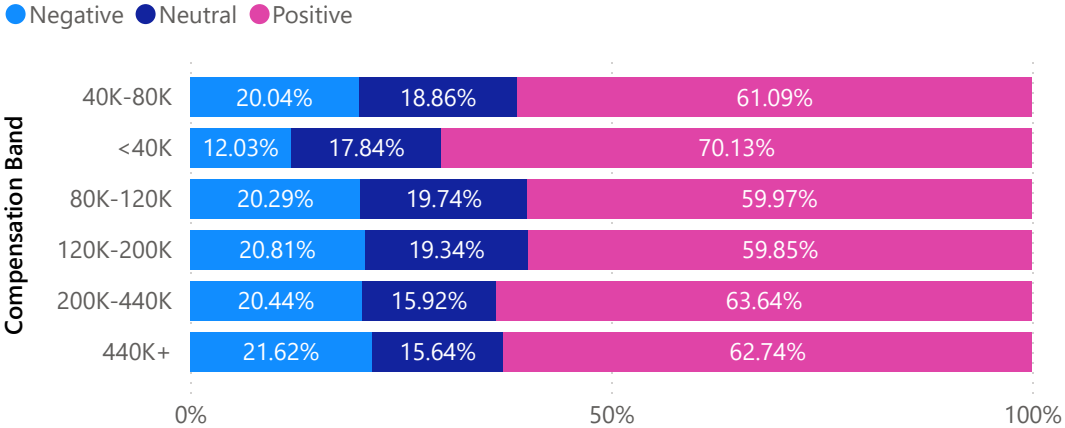
7.20

AvgJobSatisfaction

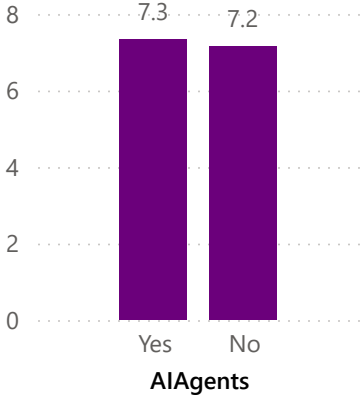
75.32K

MedianCompensation

AI Sentiment Distribution by Compensation Band



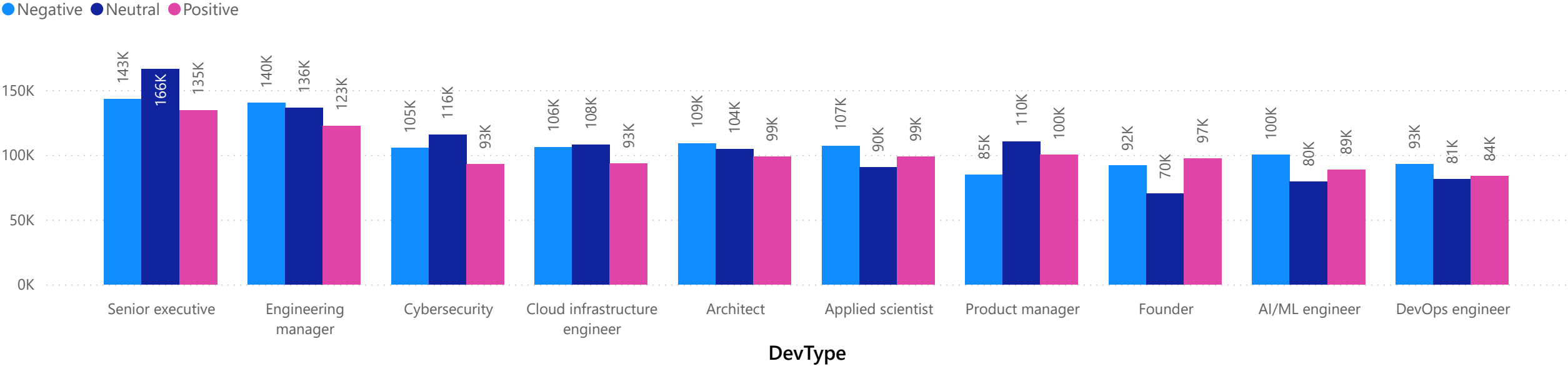
AvgJobSatisfaction by AI Agents



YearsCodingBand

- Select all
- 11-20
- 1-5
- 21+
- 6-10

MedianCompensation by DevType and AISentimentGroup



The compensation page shows a clear progression by coding experience. As YearsCodingBand moves from early-career to more experienced groups, both **median compensation** and **average job satisfaction** increase. Median compensation rises from about **16.71K** in the **1–5 years** group to **48.73K** in **6–10 years**, **81.21K** in **11–20 years**, and **105.46K** in the **21+ years** group. Job satisfaction also increases gradually, from **6.84** in the least-experienced group to **7.42** in the most-experienced group. Across compensation bands, **positive AI sentiment** remains the largest share, showing that favorable views of AI are not limited to one salary bracket. Respondents who use AI agents also report **slightly higher job satisfaction** than non-users, although the difference is modest. The role-level compensation chart further shows that **developer role and career stage** explain compensation more strongly than AI sentiment alone.

Insight: The strongest overall pattern is that experience and role drive compensation most clearly, while AI-related factors are associated more modestly with outcomes. Median compensation rises sharply with coding experience, and average job satisfaction also improves gradually across experience bands. At the same time, positive AI sentiment remains the dominant sentiment across all compensation bands, and respondents who use AI agents tend to report slightly higher job satisfaction than non-users. This suggests that AI-related attitudes and usage are connected to better outcomes, but their effect appears smaller than the structural effects of career stage and developer role.

Q4:

What programming languages and AI models are most prominent within selected industries, and how does the selected industry contribute to overall AI-model usage across developer roles?

Data Relationships Explored:

This question explores **part-to-whole**, **ranking**, and **nominal comparison** relationships.

- **Part-to-whole** is used to show how language and AI-model usage is distributed within the selected industry.
- **Ranking** is used to identify the top languages and top AI models in the selected industry.
- **Nominal comparison** is used in the matrix to compare selected-industry AI-model usage against all-industry AI-model usage across developer roles.

First Chart: Top 10 Language Share within Selected Industry

Displays the percentage share of the top programming languages within the currently selected industry.

Second Chart: Top 10 AI Model Share within Selected Industry

Displays the percentage share of the top AI models within the currently selected industry.

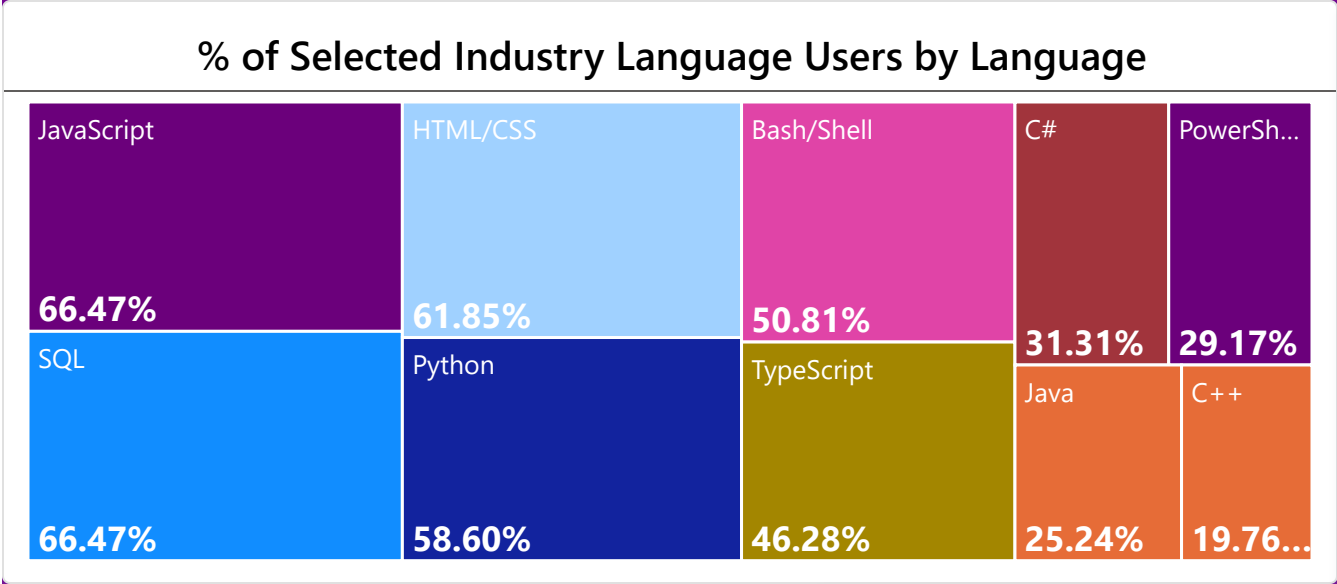
Third Chart: Developer Role Comparison of Selected Industry vs All Industries

Compares developer roles using selected-industry AI-model usage, all-industry AI-model usage, and the selected industry's share of total AI-model usage.

Slicer: Industry

Allows the page to be filtered by major industries to compare how technology usage patterns shift across sectors.

Technology Usage by Industry



AIModel	Selected Industry AIModelUsers	AIModel Rank in Selected Industry
OpenAI GPT	520	1
Anthropic Claude Sonnet	247	2
Gemini Flash	207	3
OpenAI Reasoning	205	4
OpenAI Image models	165	5
Gemini Pro	142	6
DeepSeek - Reasoning	133	7
Meta Llama	115	8
DeepSeek - General	77	9
Grok	56	10

Industry		
Select all	Government	Manufacturing
Banking/Financial Services	Healthcare	Media & Advertising Servic...
Computer Systems Design and Servic...	Higher Education	Retail and Consumer Services
Energy	Insurance	Software Development
Fintech	Internet, Telecomm or Information Se...	Transportation, or Supply Chain

DevType	Selected Industry AIModelUsers	All Industries AIModelUsers	Industry Share of All AIModel Users
Senior executive	16	292	5.48%
Engineering manager	22	495	4.44%
Developer, mobile	19	495	3.84%
Developer, full-stack	199	4943	4.03%
Developer, front-end	24	769	3.12%
Developer, desktop applications	23	580	3.97%
Developer, back-end	62	2464	2.52%
Data scientist	19	262	7.25%
Data engineer	25	316	7.91%
Architect	48	1128	4.26%
Total	457	11744	3.89%

The industry views show that the core language stack remains fairly stable across sectors, with the treemap repeatedly dominated by widely used languages such as **JavaScript, SQL, Python, HTML/CSS, Bash/Shell, TypeScript, and Java**, although their exact ordering changes by industry. The AI-model table shows an even stronger concentration pattern: **OpenAI GPT** consistently ranks first, followed by **Anthropic Claude Sonnet**, while other models such as **Gemini Flash, OpenAI Reasoning**, and **OpenAI Image models** appear at lower ranks. The role-level matrix adds the main analytical insight by showing how much each selected industry contributes to overall AI-model usage across developer roles. Most industries contribute only a small share, while sectors such as **Software Development** stand out with much stronger overall contribution. This shows that industry differences are more visible in the **scale and concentration of AI-model usage across roles** than in the leading tools themselves.

Insight: The strongest pattern is that industries use a broadly similar set of core languages and are all led by **OpenAI GPT** on the AI-model side, but they differ much more in **how strongly they contribute to overall AI-model usage across roles**. In other words, the main tools stay relatively stable, while the real differences appear in the depth of AI-model adoption and the role-level concentration within each industry.

Q5:

How does AI learning behavior connect to AI-agent usage and overall AI sentiment, and how do these patterns vary across education levels?

Data Relationships Explored:

This question explores a **flow-based categorical relationship** using an AppSource Sankey visual, together with a **nominal comparison** across education levels. The Sankey connects three stages of respondent behavior:

- whether respondents learned AI (LearnCodeAI_YesNo)
- whether they use AI agents (AIAgents)
- and their final AI sentiment group (AISentimentGroup)

The supporting matrix compares education levels using **% Learned AI**, **% Using AI Agents**, and **% Positive AI Sentiment**.

Chart 1: Sankey Flow from AI Learning to AI Usage to AI Sentiment

Displays the flow of respondents from LearnCodeAI_YesNo to AIAgents, and from AIAgents to AISentimentGroup, showing how AI learning and AI-agent usage connect to positive, neutral, and negative sentiment.

Chart 2: AI Learning, AI-Agent Use, and Positive Sentiment by Education Level

Compares education levels using % Learned AI, % Using AI Agents, and % Positive AI Sentiment to show whether AI engagement differs across educational backgrounds.

Note:

Sankey report excludes No Response values and keeps only valid Yes/No and sentiment categories

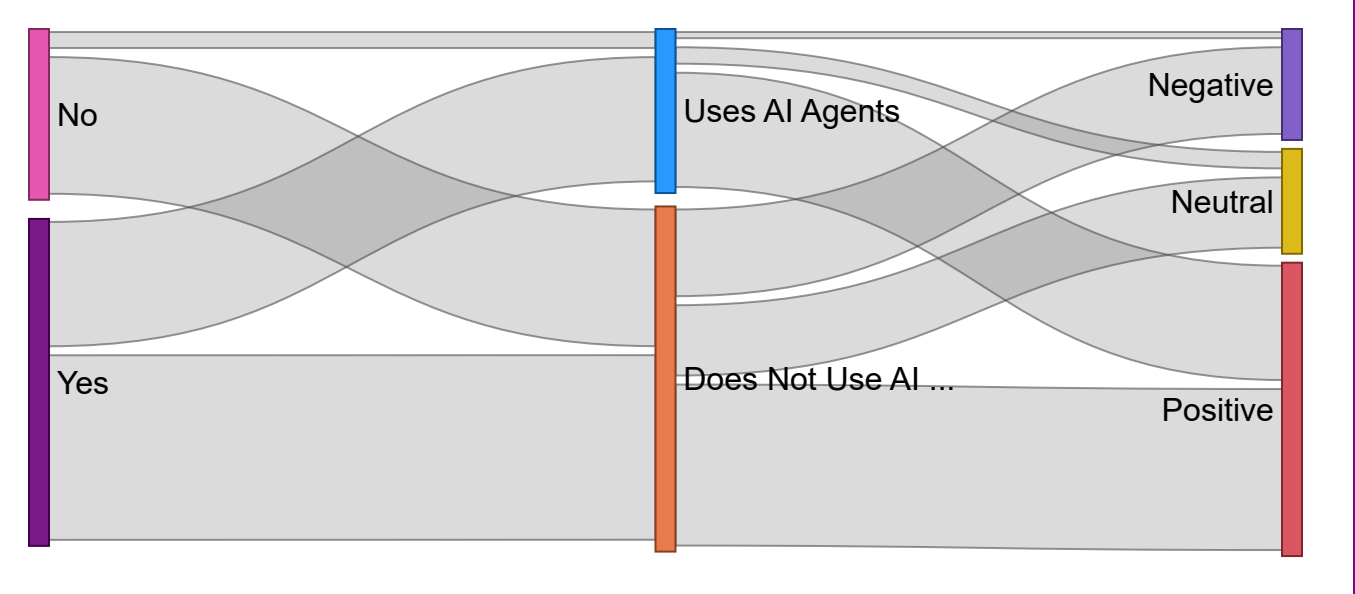
AI Learning and Sentiment Flow

EdLevel	% Learned AI	% Using AI Agents	% Positive AI Sentiment
Associate degree	0.67	0.30	0.58
Bachelor's degree	0.69	0.32	0.62
Master's degree	0.70	0.32	0.62
Non degree	0.64	0.31	0.56
Primary school	0.60	0.34	0.59
Professional degree	0.67	0.27	0.57
Secondary school	0.56	0.24	0.50
Total	0.68	0.31	0.60

Learned Coding AI

Used AI Agents

Sentiment Towards AI



The Sankey chart shows a clear progression from **AI learning** to **AI-agent usage** and then to **overall AI sentiment**. Respondents who learned coding AI are not necessarily **Uses AI Agents** which is expected as AI Agents are advanced tools and most of the work can be done with familiarity with AI , while respondents who did not learn coding AI flow more strongly into the **Does Not Use AI Agents** group. The second part of the flow shows that respondents who use AI agents move more strongly into the **Positive** sentiment group, while non-users show a more mixed distribution across **Positive**, **Neutral**, and **Negative** sentiment. This suggests that direct AI exposure is associated with more favorable views toward AI.

The education-level matrix supports this same general pattern. Learning AI rates are fairly high across most education groups, with **master's degree holders (0.70)** and **bachelor's degree holders (0.69)** slightly above the overall average (**0.68**). AI-agent usage is lower than AI learning across all groups, but remains relatively strongest for **primary school (0.34)**, **bachelor's degree (0.32)**, and **master's degree (0.32)** compared with the overall rate (**0.31**). Positive AI sentiment is also highest among **bachelor's** and **master's degree** holders (**0.62**), while **secondary school (0.50)** and **non-degree (0.56)** groups are somewhat lower. Overall, the matrix suggests that stronger formal education is associated with slightly stronger AI learning and more positive sentiment, although the differences are moderate rather than extreme.

Insight: The strongest pattern is that **AI learning, AI-agent usage, and positive AI sentiment are connected in sequence**, and this pattern is also supported by education-level differences. Respondents who learn AI are not necessarily to use AI agents, and respondents who use AI agents are more likely to report positive sentiment toward AI. The education matrix further suggests that respondents with **bachelor's and master's degrees** tend to show slightly stronger AI learning and more positive sentiment than some lower education groups, reinforcing the idea that greater exposure and readiness are linked to more favorable AI attitudes.

Additional Information

AppSource Visual

An AppSource **Sankey** visual was used to show the flow from **AI Learning** → **AI-Agent Usage** → **AI Sentiment**. To support this visual, a separate flow table was created by filtering valid category values, grouping them into flow counts, and appending them into a single Source–Destination–Value structure.

Advanced DAX Measures

Several DAX measures were created to support the report, including TotalRespondents, AvgWorkExperience, AvgYearsCoding, AvgJobSatisfaction, MedianCompensation, % Learned AI, % Using AI Agents, and % Positive AI Sentiment. These measures were used in KPI cards, percentage-based visuals, and cross-group comparison.

Context Manipulation

Context manipulation was used on the **Technology Usage by Industry** report to compare **selected-industry AI-model usage** against **all-industry AI-model usage**. This was implemented by removing the Industry filter in the denominator while keeping the current developer-role context active, allowing the report to show how much the selected industry contributes to the overall AI-model usage of each role.

Context Manipulation DAX Example

All Industries AIModelUsers = CALCULATE([AIModelUsers], ALL(MainRespondents[Industry]))

Industry Share of All AIModel Users = DIVIDE([Selected Industry AIModelUsers], [All Industries AIModelUsers], 0)

Extra DAX Function

The DAX function **RANKX** was used to rank AI models within the selected industry based on Selected Industry AIModelUsers. This helped identify the relative position of each AI model under the current industry filter context.

Extra DAX Example

AIModel Rank in Selected Industry = RANKX(ALLSELECTED(AIModels[AIModel]), [Selected Industry AIModelUsers], , DESC)